

WAF T SYSTEM OVERVIEW

Complete Architecture Documentation

What is WAF T?

WAF T (Worldbuilding & AI Framework for Teleport) is a comprehensive Python framework for:

Evolutionary AI

Breed AI agents through directed mutation and selection

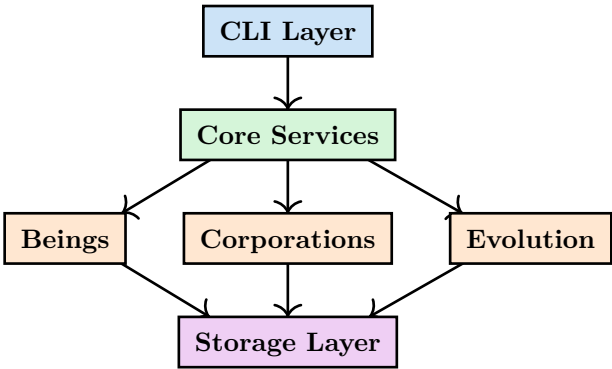
Worldbuilding

Create and simulate corporations, beings, and realities

Documentation

Generate professional PDFs, reports, and templates

High-Level Architecture



Core Components

1. CLI Layer (src/waft/cli/)

Command-line interface for all WAF T operations:

- waft new — Create laboratories
- waft evolve — Run evolution cycles
- waft spawn — Create beings/agents
- waft status — Check system state

2. Core Services (src/waft/core/)

The heart of WAF T with 169 Python modules:

Module	Purpose
being.py	Timeful agents with skills, memories, goals
corporations/	Corporate simulation (Teleport Massive)
evolution/	Agent breeding and mutation

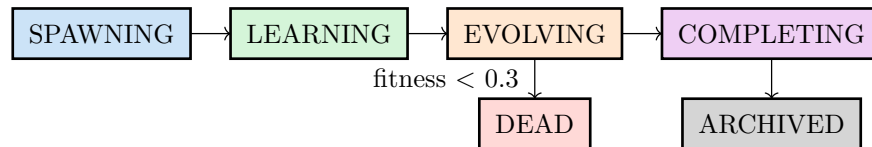
<code>chronicler/</code>	Automated documentation generation
<code>empirica.py</code>	Epistemic self-assessment
<code>science/</code>	Research and observation framework

3. Beings System (src/waft/being.py)

Beings are **timeful, dynamic entities** that exist in realities:

```
class Being:
    being_id: str          # Unique identifier
    reality_id: str        # Which reality they exist in
    skills: dict[str, float] # Learned abilities (0-10)
    memories: list         # Experiences
    lessons: list          # What worked/didn't work
    fitness: float          # Evolutionary success
    personality: dict      # Traits
    goals: list            # Objectives
    state: BeingState      # SPAWNING, LEARNING, EVOLVING, etc.
```

Being States



4. Corporations System (src/waft/core/corporations/)

Full economic simulation:

Corporation

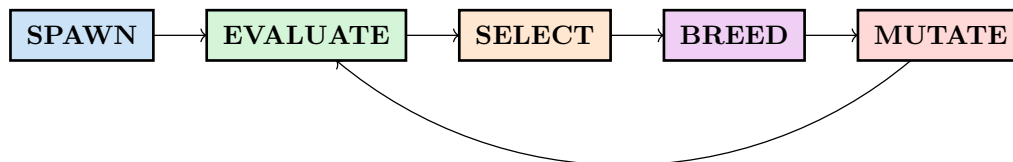
- Financial state tracking
- Department management
- Employee (Being) roster
- Economic transactions

Simulation

- Time advancement
- Monthly expenses
- Revenue tracking
- Runway calculation

5. Evolution System (src/waft/evolution/)

Directed evolution of agents:



Module Count by Area

Area	Files	Description
core/	169	Core business logic
templates/	182	Document templates + fonts
evolution/	30	Evolution engine
pantheon/	20	Entity gods system
api/	22	REST API routes
ui/	13	Dashboard and UI
ai_town/	7	Multi-agent town simulation
Total	443+	Complete framework

Key Integrations

Empirica

Epistemic self-assessment:

- 13 knowledge vectors
- Preflight/Postflight assessments
- Gamification (XP, Karma, Levels)

Typst

Document generation:

- Professional PDFs
- Multiple template systems
- Automatic compilation

LLM

AI integration:

- OpenAI compatible
- Local model support
- Prompt engineering

D&D 5e

RPG mechanics:

- Character sheets
- Combat system
- Scenario generation

Summary Statistics

